Linear Algebra

Vectors, matrices, and dot products

Worked answers

Name _____ Date _____

Choose one answer per question · Use the margins for calculations and sketches · 25 minutes

01 Vectors and Span

1 Why should we understand **how space changes**, as well as calculate with matrices?

(1) A diagram always lets us skip exact numerical calculations
(2) It helps us interpret a calculation and decide which operation to use
(3) It shows that every matrix performs the same rotation
(4) Geometry is unrelated to calculation and can be memorized separately

Answer (2) Calculations give a result; geometry helps explain what the operation means and when to use it

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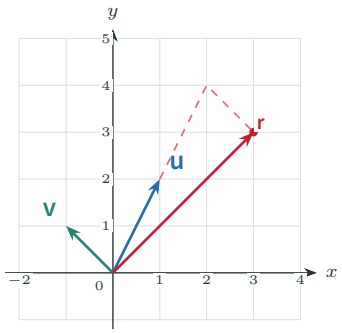
2 **Calculate and draw $\mathbf{r} = 2\mathbf{u} - \mathbf{v}$ on the grid** Choose its coordinates

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

(1) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (2) $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$

(3) $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ (4) $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$



Answer (3) $2\mathbf{u} - \mathbf{v} = (2, 4) - (-1, 1) = (3, 3)$
Subtracting a negative is the same as adding a positive

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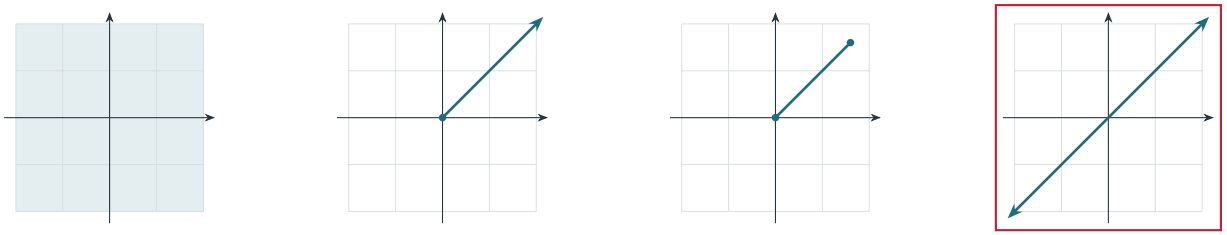
3 As a, b range over **all real numbers**, which set contains every possible vector endpoint?

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \text{Span}(\mathbf{u}, \mathbf{v}) = \{a\mathbf{u} + b\mathbf{v} \mid a, b \in \mathbb{R}\}$$



(1) The entire plane (2) A positive ray (3) A line segment (4) An infinite line

Answer (4) $\mathbf{v} = 2\mathbf{u}$, so $a\mathbf{u} + b\mathbf{v} = (a + 2b)\mathbf{u}$
Negative multiples are allowed, giving the whole line $y = x$ through the origin in both directions

02 Basis and Linear Transformations

4 Which set of vectors can be a **basis for the entire plane**?

(1) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}$ (2) $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\4\end{bmatrix}$

(3) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\0\end{bmatrix}$ (4) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}$

Answer (1) The two nonparallel vectors span the plane without redundancy
In (4), the third vector is the sum of the first two: the set spans the plane but is not a basis

5 Which is a **linear transformation**?

$T(a\mathbf{u}+b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$

(1) $T(x,y) = (x+1,\ y)$ (2) $T(x,y) = (x^2,\ y)$

(3) $T(x,y) = (2x,\ x+y)$ (4) $T(x,y) = (x,\ |y|)$

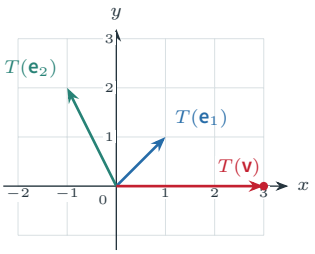
Answer (3) It preserves addition and multiplication by a real number
(1) moves the origin; in (2), doubling x quadruples x^2
(4) has $T(0,-1) \neq -T(0,1)$, so it does not preserve negative multiples

6 A linear transformation T moves the two basis vectors as shown Choose **its matrix** A

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-1\\2\end{bmatrix}$$

(1) $\begin{bmatrix}1 & 1\\-1 & 2\end{bmatrix}$ (2) $\begin{bmatrix}1 & -1\\1 & 2\end{bmatrix}$

(3) $\begin{bmatrix}-1 & 1\\2 & 1\end{bmatrix}$ (4) $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$



Answer (2) The first column is $T(\mathbf{e}_1)$; the second is $T(\mathbf{e}_2)$ $\mathbf{e}_1 = (1,0),\ \mathbf{e}_2 = (0,1)$

7 Apply T from Question 6 to $\mathbf{v}$ **Plot** $T(\mathbf{v})$ **on the grid above** and choose its coordinates

$$\mathbf{v} = \begin{bmatrix}2\\-1\end{bmatrix}, \qquad T(\mathbf{v}) = 2T(\mathbf{e}_1) - T(\mathbf{e}_2)$$

(1) $\begin{bmatrix}1\\-4\end{bmatrix}$ (2) $\begin{bmatrix}1\\0\end{bmatrix}$

(3) $\begin{bmatrix}3\\4\end{bmatrix}$ (4) $\begin{bmatrix}3\\0\end{bmatrix}$

Answer (4) $T(\mathbf{v}) = 2(1,1) - (-1,2) = (3,0)$
Use the same original coefficients $2, -1$ with the transformed basis vectors

03 Composition and Dot Products

8 Apply A **first, then** B to **x** Choose the correct combined matrix and result

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(1) BA, result $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

(2) AB, result $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

(3) BA, result $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(4) AB, result $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\mathbf{x} \xrightarrow{A} \underbrace{\qquad}_{\text{intermediate}} \xrightarrow{B} \underbrace{\qquad}_{\text{final result}}$$

Answer (1) $A\mathbf{x} = (2, 1)$, $B(A\mathbf{x}) = (-1, 2)$ The combined matrix is BA
Reversing the order gives $A(B\mathbf{x}) = (0, 1)$, a different result

9 **Draw a perpendicular** from the tip of **w** to the line through **v** Choose the projection length and dot product

$$\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

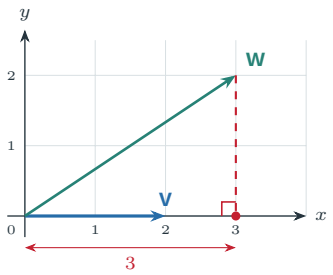
$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2$

(1) Projection length 3, dot product 3

(2) Projection length 2, dot product 4

(3) Projection length 3, dot product 6

(4) Projection length 2, dot product 6



Answer (3) Projection length 3; $\mathbf{v} \cdot \mathbf{w} = 2 \times 3 + 0 \times 2 = 6$
Here the dot product is the projection length times $\|\mathbf{v}\| = 2$, not the projection length alone

10 A linear map L sends **a vector to one number** For $L(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x}$, choose the correct **w** and result

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 2, \qquad L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = -1$$

(1) $\mathbf{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, result -1

(2) $\mathbf{w} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, result 5

(3) $\mathbf{w} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, result $\begin{bmatrix} 6 \\ -1 \end{bmatrix}$

(4) $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, result 7

Answer (2) $\mathbf{w} = (2, -1)$ and $L(\mathbf{x}) = 2x_1 - x_2$, so $L(3, 1) = 6 - 1 = 5$
Duality here means that one vector **w** represents a linear map to a number
In ML, $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x}$ with $b = 0$ has the same form

Scope Six 3Blue1Brown videos · Questions and diagrams created separately

Visual intuition Vectors Linear combinations, span, basis Transformations and matrices Composition Dot products and duality

Content reference Official introduction transcript Vectors Span Transformations Composition Dot Products